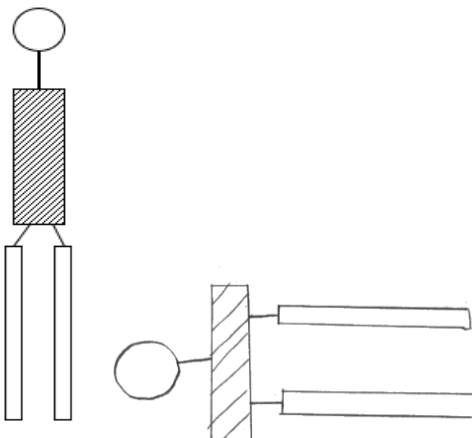


Question number	Answer	Additional guidance	Mark
7(a)(i)	A diagram that includes the following points: • one glycerol (shaded / labelled), two fatty acids (not shaded / labelled), one phosphate and three bonds shown (1) • all components drawn together correctly (1)	 ecf if: {one / three} fatty acids drawn attached to glycerol but rest correct - glycerol missing but fatty acids attached to head with bonds - bonds missing but all four components are touching correctly {all shaded / nothing shaded} with no labels	(2)

Question number	Answer	Mark
7(a)(ii)	The only correct answer is C, $\begin{array}{c} & & O \\ -C- & O- & C- \\ & & \end{array}$ <i>A is incorrect because it is not an ester bond</i> <i>B is incorrect because it is not an ester bond</i> <i>D is incorrect because it is not an ester bond</i>	(1)

Question number	Answer	Additional guidance	Mark
7(b)(i)	An explanation that includes two of the following points: - because {phospholipids / molecules} have more (heat / kinetic) energy (1) - because {forces / interactions} between {fatty acid chains / phospholipids} break (1) - therefore <u>phospholipids</u> can move around (within the membrane) / phospholipid bilayer} more (1)	IGNORE proteins ACCEPT bonds break bonds between {fatty acids and cholesterol / phospholipids} Van der waals forces	(2)

Question number	Answer	Mark				
7(b)(ii)	The only correct answer is C <table><tr><th>Has double bonds between two carbons</th><th>Number of hydrogens</th></tr><tr><td>yes</td><td>fewer than the saturated fatty acid</td></tr></table> <i>A is incorrect because unsaturated fatty acids have double bonds between carbons</i> <i>B is incorrect because unsaturated fatty acids have double bonds between carbons</i> <i>D is incorrect because unsaturated fatty acids have fewer hydrogen</i>	Has double bonds between two carbons	Number of hydrogens	yes	fewer than the saturated fatty acid	(1)
Has double bonds between two carbons	Number of hydrogens					
yes	fewer than the saturated fatty acid					

Question number	Answer	Additional guidance	Mark
7(b)(iii)	An explanation that includes the following points: - cholesterol interacts with {phospholipids / fatty acids} (1) (if less cholesterol) movement of <u>phospholipids</u> no longer restricted (by interaction with cholesterol) (1)	NB we are not looking for the idea that cholesterol is acting as a barrier or taking up space ACCEPT bonds ACCEPT so <u>phospholipids</u> can move around more	(2)

Question number	Answer	Additional guidance	Mark
7(b)(iv)	will require less (kinetic) energy to move / because there are fewer interactions (with {other fatty acids / phospholipids / cholesterol}) (1)	ACCEPT bonds converse for longer chains	(1)

Question number	Answer	Additional guidance	Mark
7(c)(i)	An answer that includes the following points: - convert saturated fatty acids into unsaturated fatty acids (1) - convert long side chains into shorter ones (1)	ACCEPT become unsaturated / increase in unsaturated / decrease in saturated fats ACCEPT become shorter	(2)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	An answer that includes two of the following points: • to maintain the fluidity of the membrane (1) • so that membrane can change shape (1) • so that fusion of membranes can occur (1) • so that substances can continue to move across membrane (1)	ACCEPT otherwise the fluidity will decrease ACCEPT named example e.g phagocytosis, endocytosis, exocytosis, red blood cells, formation of vesicles ACCEPT named example e.g. of nuclei (during fertilisation) fusion of vesicles ACCEPT named substance	(2)